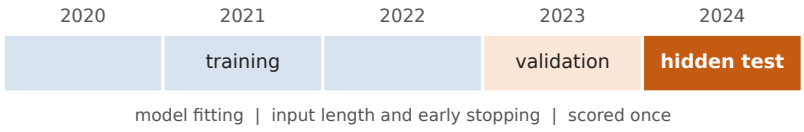
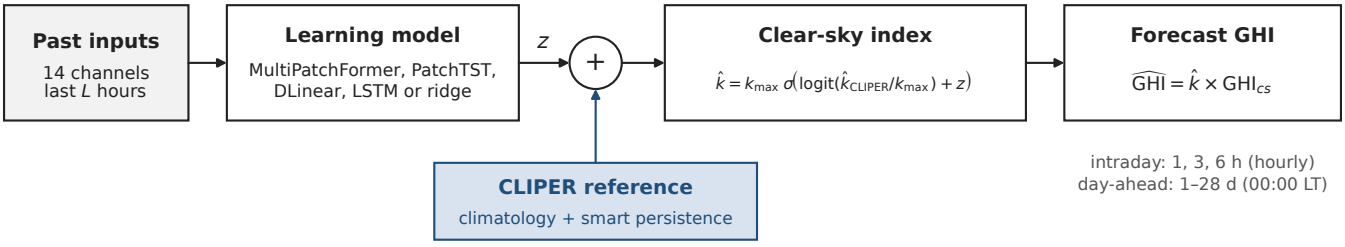
(a) Data and chronological split

Solcast estimated actuals (satellite-derived), hourly
four cities: Islamabad, Karachi, Lahore, Multan



(b) Forecasting chain (all models, all horizons)



(c) Evaluation on 2024

Verification
RMSE, MAE, MBE, skill vs CLIPER
DM tests (BH-adjusted), 95% intervals

Robustness and operations
3 seeds; ablation; blind city;
missing input; longer training

PV assessment (separate)
plane-of-array yield, payback scenarios
forecast error → energy error